Analysis and Design of Algorithms

NUS National University of Singapore

Algorithms

CS3230

Tutorial

Week 9

Question 1

PARTITION: Given an array of positive integers $A[1..n]$, can the elements in the array be partitioned into two sets of equal total sum? Consider the following algorithm, which answers the question of whether a subset of $A[1..k]$ that sums to `total` exists.

```
Partition(A,k,total)

if total = 0 then return true

if k = 0 then return false

return (Partition(A,k-1,total) or Partition(A,k-1,total-A[k]))
```

To answer the question of whether the set can be partitioned into two equal subsets, we can first sum the elements in A as S and then run `Partition(A,n,S/2)` if S is even, and return `false` if S is odd.

What is the runtime of the algorithm?

- $\Theta(n^2)$
- $\Theta(n)$
- $\Theta(S)$
- $\Theta(Sn)$
- $\Theta(2^n)$

Question 1

Answer: E $\Theta(2^n)$

First look at correctness (***optimal substructure***). Assume an optimal solution consisting of a subset *U* that sums to *total*.

- If the solution contains $A[n]$, then the set $U-\{A[n]\}$ must sum to $total-A[n]$.
- If the solution does not contain $A[n]$ then the set $U-\{A[n]\}$ must sum to *total*.

Hence (Partition(A,n-1,*total*) **or** Partition(A,n-1,$total-A[n]$)) returns correctly.

Question 1

Answer: E $\Theta(2^n)$

The recursive algorithm computes whether there is a subset of the first *k* numbers that sums to *total*.

Consider the case where the set *cannot* be partitioned equally into two sets -- *total* is never 0. Can ignore *total* in runtime computation. Recursion tree is complete tree of height *n*, or

$T(n) = 2T(n-1) + 1 = 4T(n-1) + 2 + 1 = \ldots = 2^n T(0) + 2^{n-1} + \ldots + 1 = \Theta(2^n)$

Question 2

PARTITION: Given an array of positive integers $A[1..n]$, can the elements in the array be partitioned into two sets of equal total sum? Consider the following algorithm, which answers the question of whether a subset of $A[1..k]$ that sums to `total` exists.

We first sum the elements in A as S. If S is odd we return false. Otherwise, we initialize another array $M[1..n][0..S/2]$ with *unknown* and run the following:

```
Partition(A,k,total)

if total = 0 then return true

if k = 0 or total < 0 then return false

if M[k][total] = unknown

    M[k][total]← (Partition(A,k-1,total) or Partition(A,k-1,total-A[k]))

return M[k][total]
```

What is the runtime of the algorithm?

- $\Theta(n)$
- $\Theta(S)$
- $\Theta(Sn)$
- $\Theta(2^n)$

Question 2

Answer: C $\Theta(Sn)$

Note: This is a **memoization** algorithm.

For memoization, the runtime is usually proportional to the number of subproblems times the amount of work in each subproblem.

Question 2

Answer: C $\Theta(Sn)$

We use $\Theta(Sn)$ time to initialise the matrix.
There are $\leq Sn$ subproblems and the amount of work in each subproblem is $O(1)$.
Hence total runtime is $\Theta(Sn)$. Not $\Theta(2^n)$ because of ***overlapping subproblems***.

For memoization, the runtime is usually proportional to the number of subproblems times the amount of work in each subproblem.

Question 3

PARTITION: Given an array of positive integers $A[1..n]$, can the elements in the array be partitioned into two sets of equal total sum? Consider the following algorithm:

We first sum the elements in A as S. If S is odd we return false. Otherwise, we create another array $M[0..n][0..S/2]$, initialize $M[0][0]$ to **true** and $M[0][j]$ to **false** for $j = 1, \ldots, total$, and run the following:

```
Partition(A,total)

FIRST FOR LOOP

    SECOND FOR LOOP

        if (j ≥ A[i]) then

            M[i][j] ← M[i-1][j] or M[i-1][j-A[i]]

        else

            M[i][j] ← M[i-1][j]
```

Which of the following options for

```
FIRST FOR LOOP

    SECOND FOR LOOP
```

will give correct algorithms?

[**Hint:** Draw a dependency graph - what does **M[i][j]** depend on?]

```
(I)   for i ← 1 to n
          for j ← 0 to total
(II)  for i ← 1 to n
          for j ← total downto 0
(III) for j ← 0 to total
          for i ← 1 to n
(IV)  for j ← total downto 0
          for i ← 1 to n
```

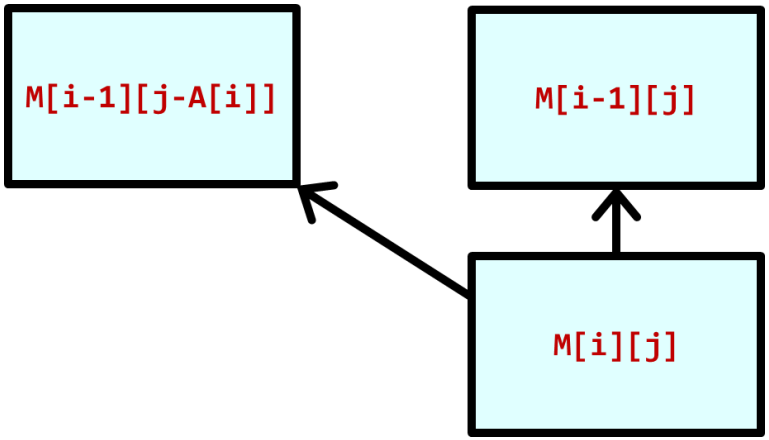
- (I) only
- (I) and (II) only
- (I), (II) and (III) only
- (I), (II), (III), and (IV)

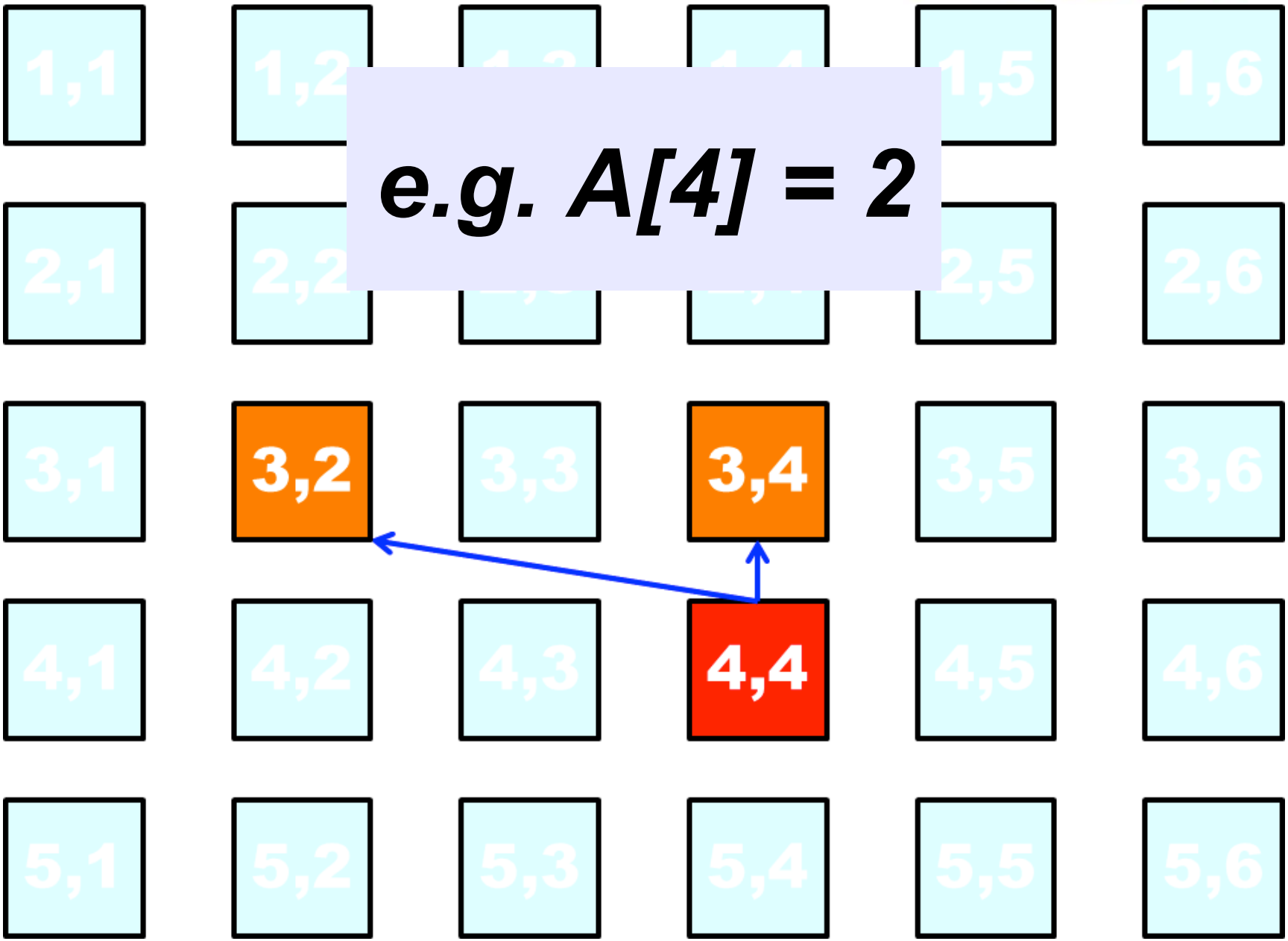
Question 3

Answer: C

This is the dynamic programming version of the algorithm -- previously recursion and memoization versions.

Ordering important in DP: ***table entries must have been computed before being used***.

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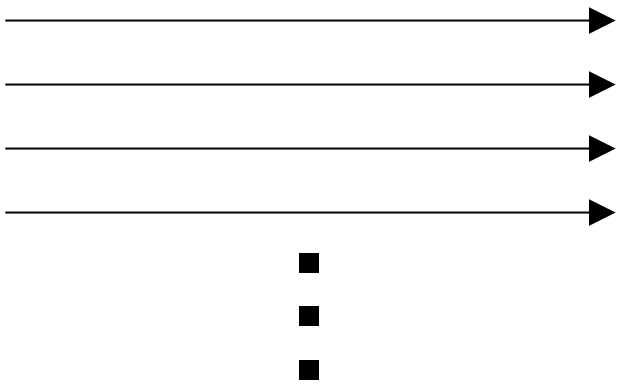
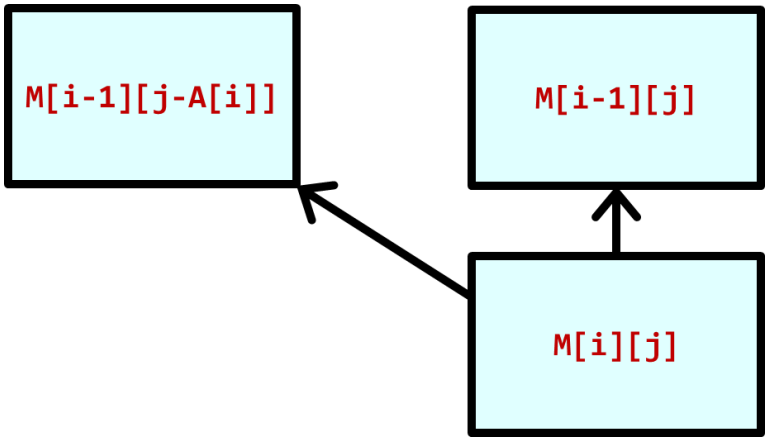
e.g. A[4] = 2

Question 3

Ordering (I): Okay

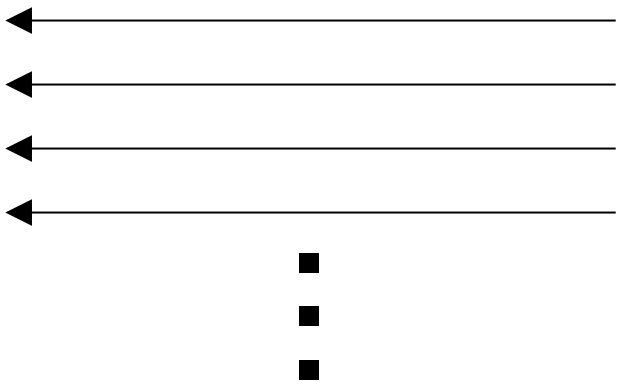


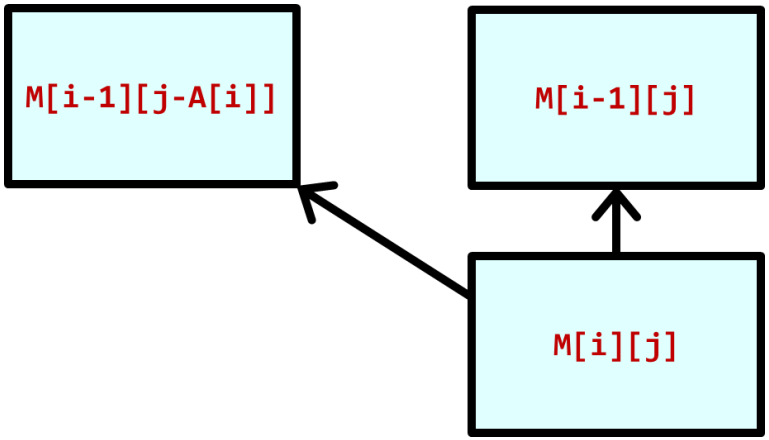



```
for i ← 1 to n
    for j ← 0 to total
```

Question 3

Ordering (II): Okay

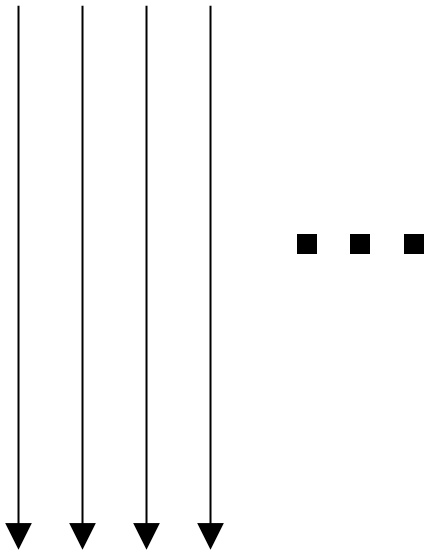





```
for i ← 1 to n
    for j ← total downto 0
```

Question 3

Ordering (III): Okay



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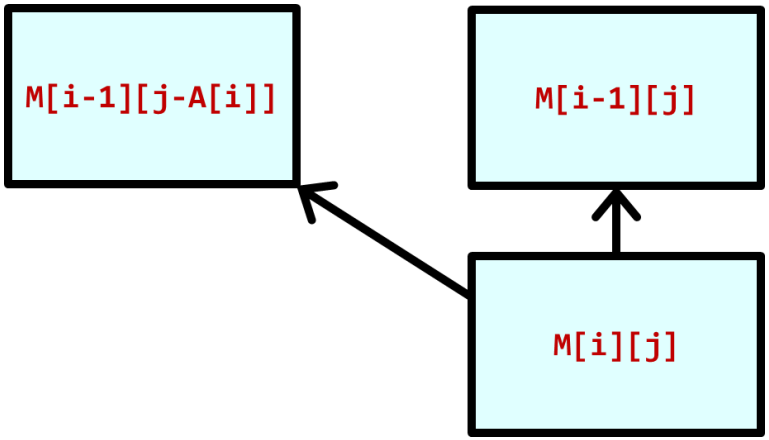
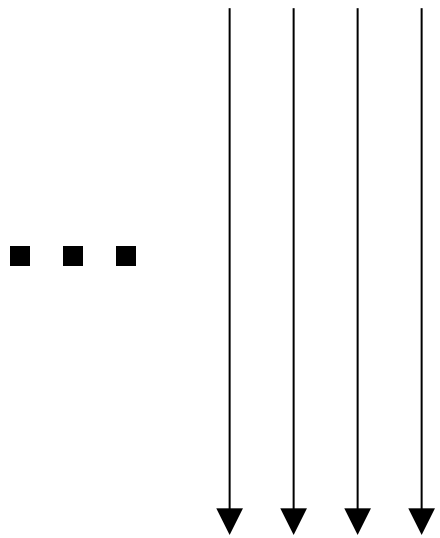
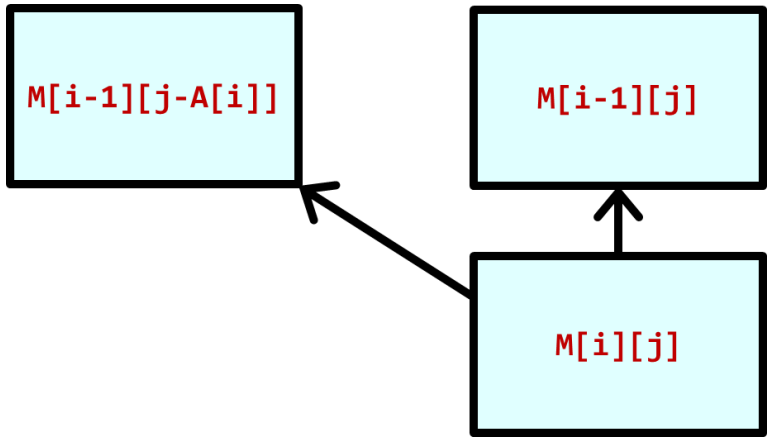


for j ← 0 to total
    for i ← 1 to n

Question 3

Ordering (IV): Wrong, table entries used before computed





for j ← total downto 0
    for i ← 1 to n

Question 4

PARTITION: Given an array of positive integers $A[1..n]$, can the elements in the array be partitioned into two sets of equal total sum?

Let S be the sum of elements in A. Which of the following statements regarding PARTITION are true?

(I) Memoization may solve fewer subproblems than bottom-up dynamic programming to obtain the final solution.

(II) It is possible to solve PARTITION using bottom-up dynamic programming in $O(nS)$ time using **only** $O(S)$ **space**.

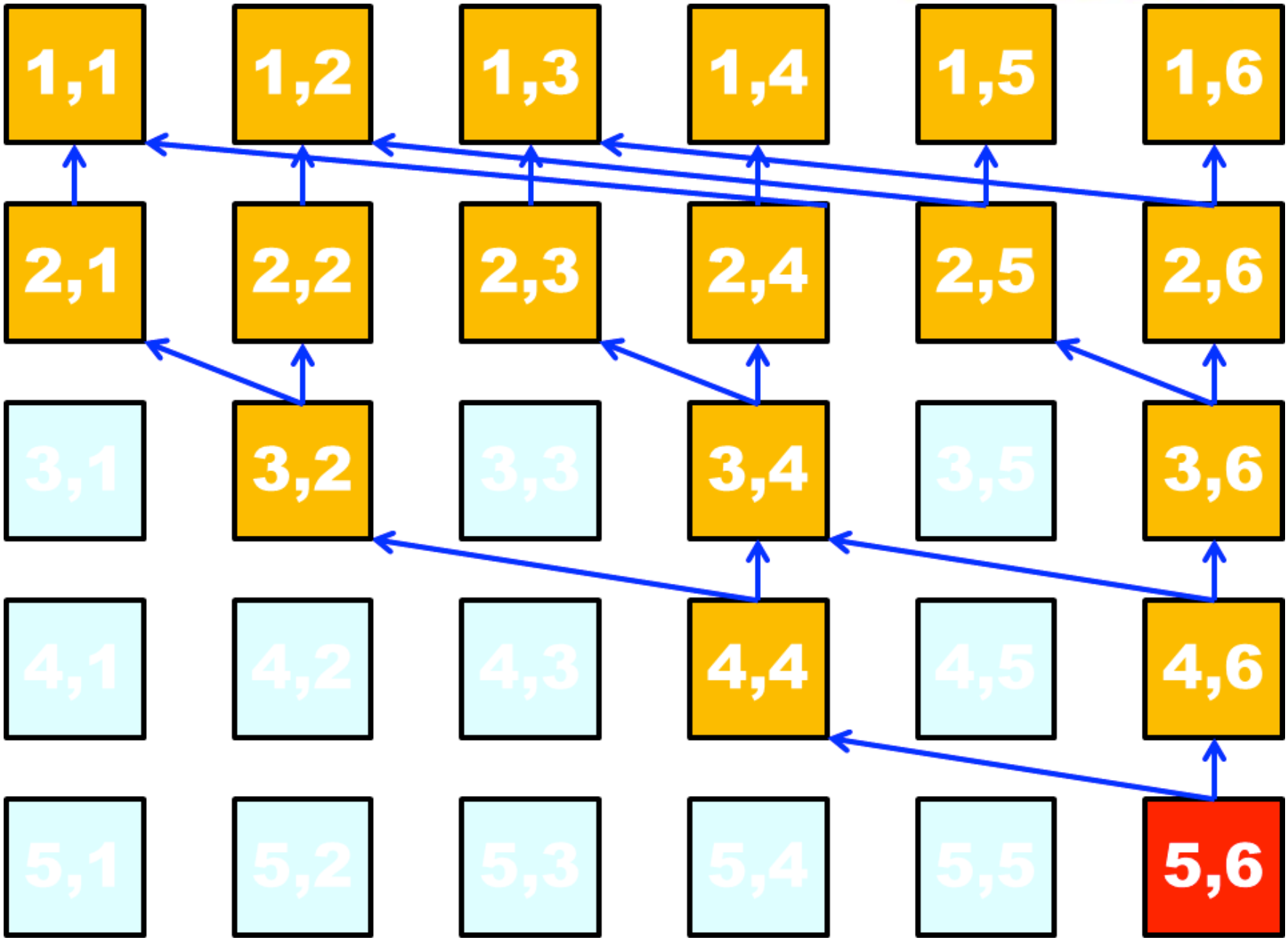
- None.
- (I) only
- (II) only
- Both (I) and (II)

Question 4

Answer: D Both (I) and (II)

For (I), memoization recursively calls (Partition(A,n-1,$total$) **or** Partition(A,n-1,$total$-A[n])).

- Only some subproblems are solved, e.g. does not solve for Partition(A,n-1,$total$-1) if A[n] is not equal to 1.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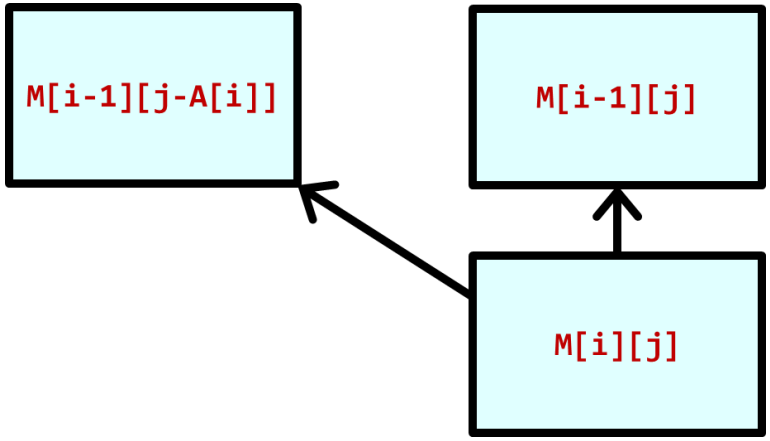
Question 4



Answer: D Both (I) and (II)

For (II): Use $O(S)$ space

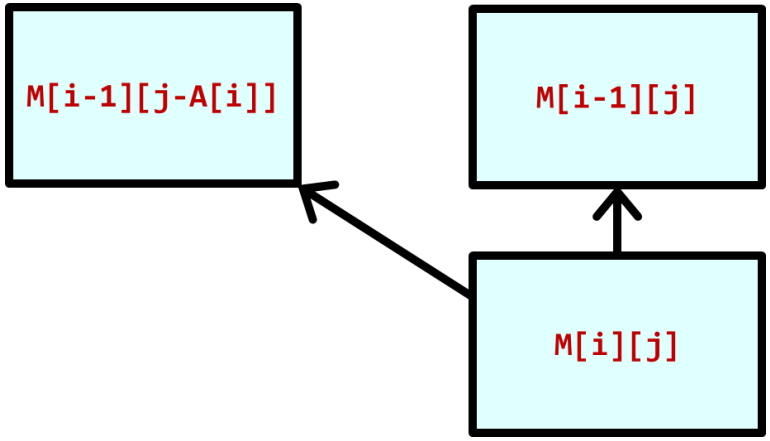
Note only previous row required to compute current row.

Store only two rows: *curr* and *prev*.

After finish computing *curr*, swap row with *prev*.

```
Partition(A,total)
for i ← 1 to n
    for j ← 0 to total
        if (j ≥ A[i]) then
            M[i][j] ← (M[i-1][j]
                  or M[i-1][j-A[i]])
        else
            M[i][j] ← M[i-1][j]
```

Question 4

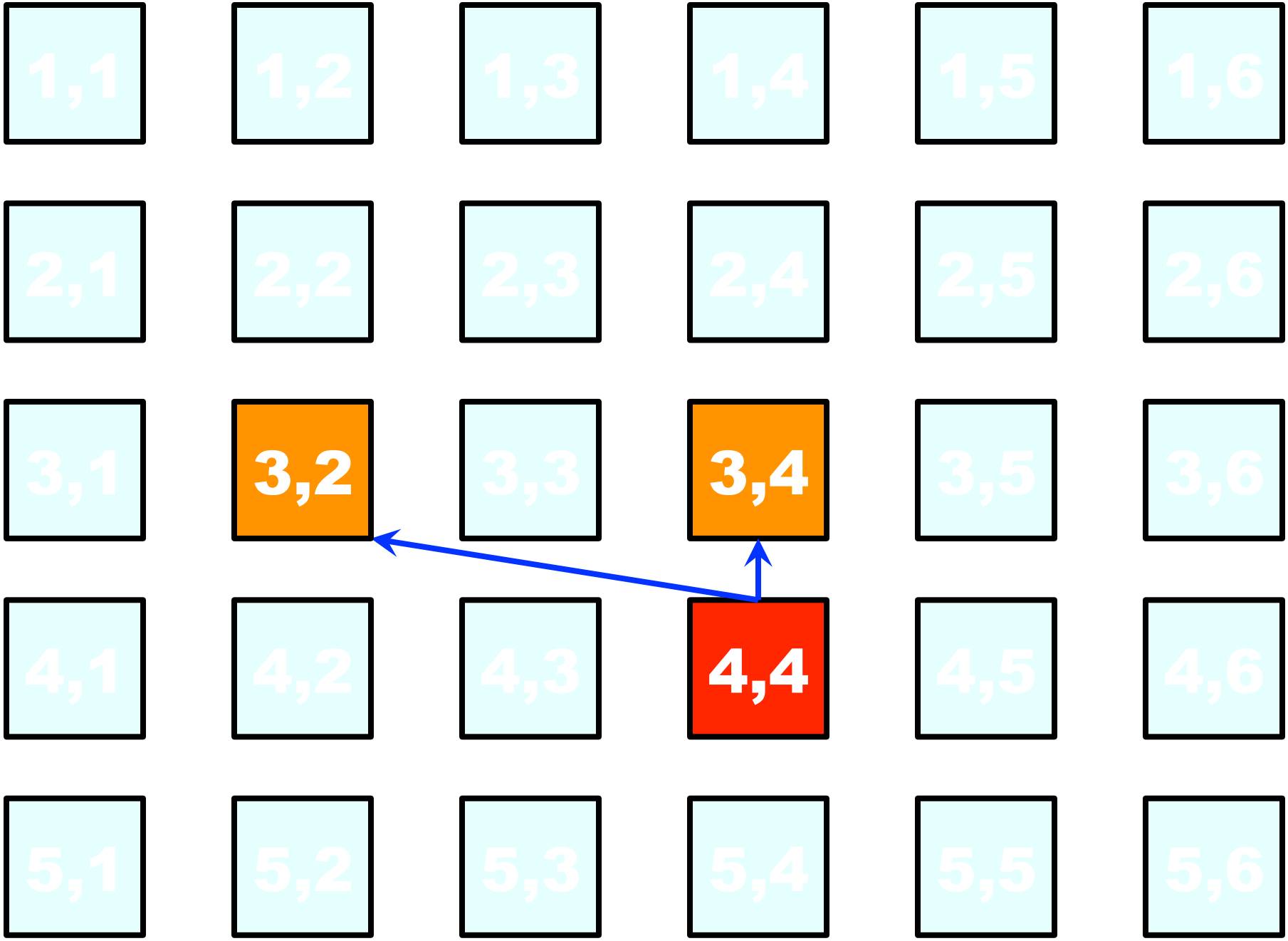


Answer: D Both (I) and (II)

For (II): Use $O(S)$ space

Even better, can use just one row only and overwrite it by iterating from *total* downto 0, as shown in code here

```
Partition(A,total)
for i ← 1 to n
    for j ← total downto 0
        if (j ≥ A[i]) then
            M[j] ← (M[j]
                    or M[j-A[i]])
        else
            M[j] ← M[j]
```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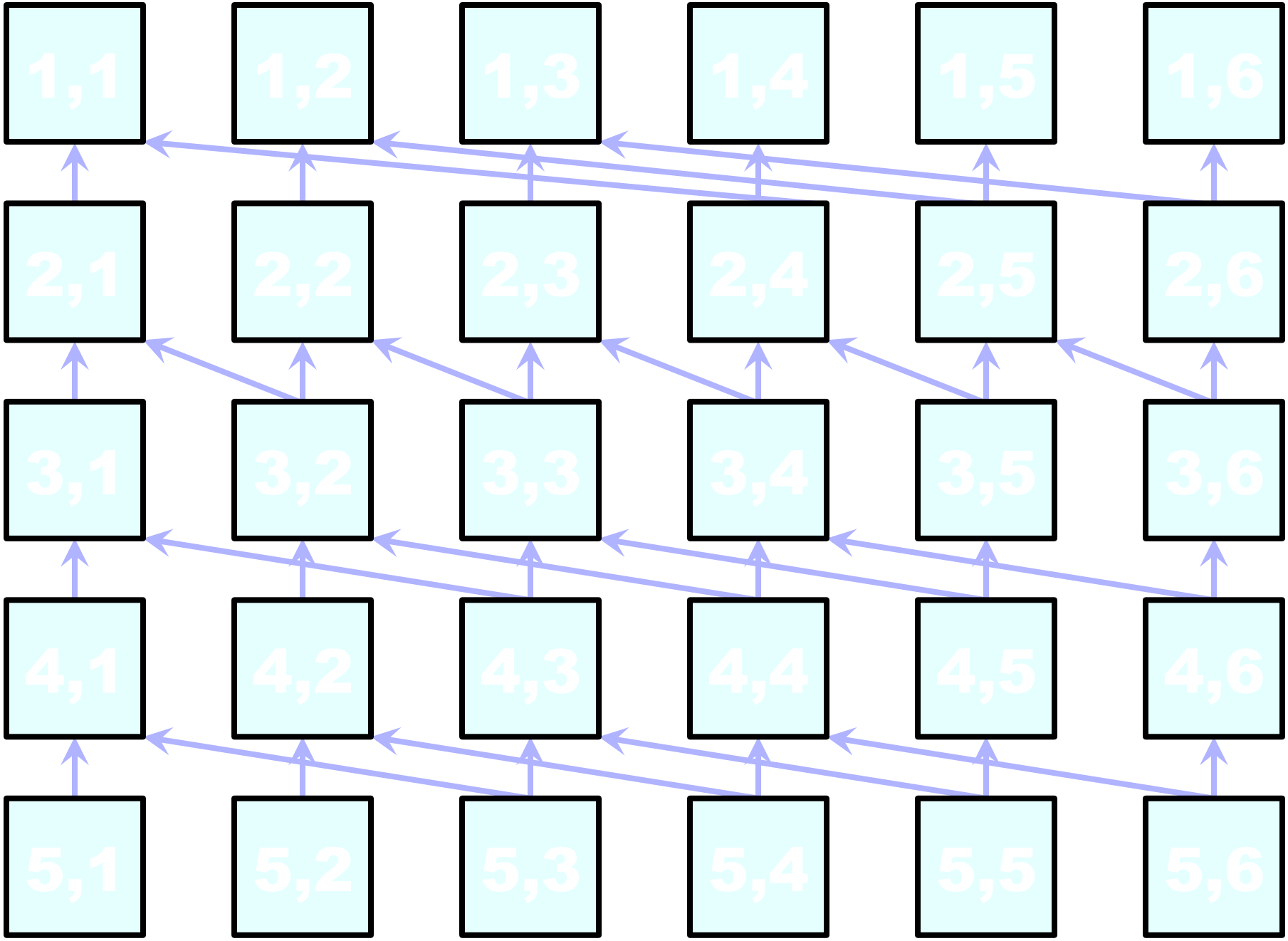
Question 4

Bottom-Up DP for PARTITION: Original $\Theta(Sn)$ space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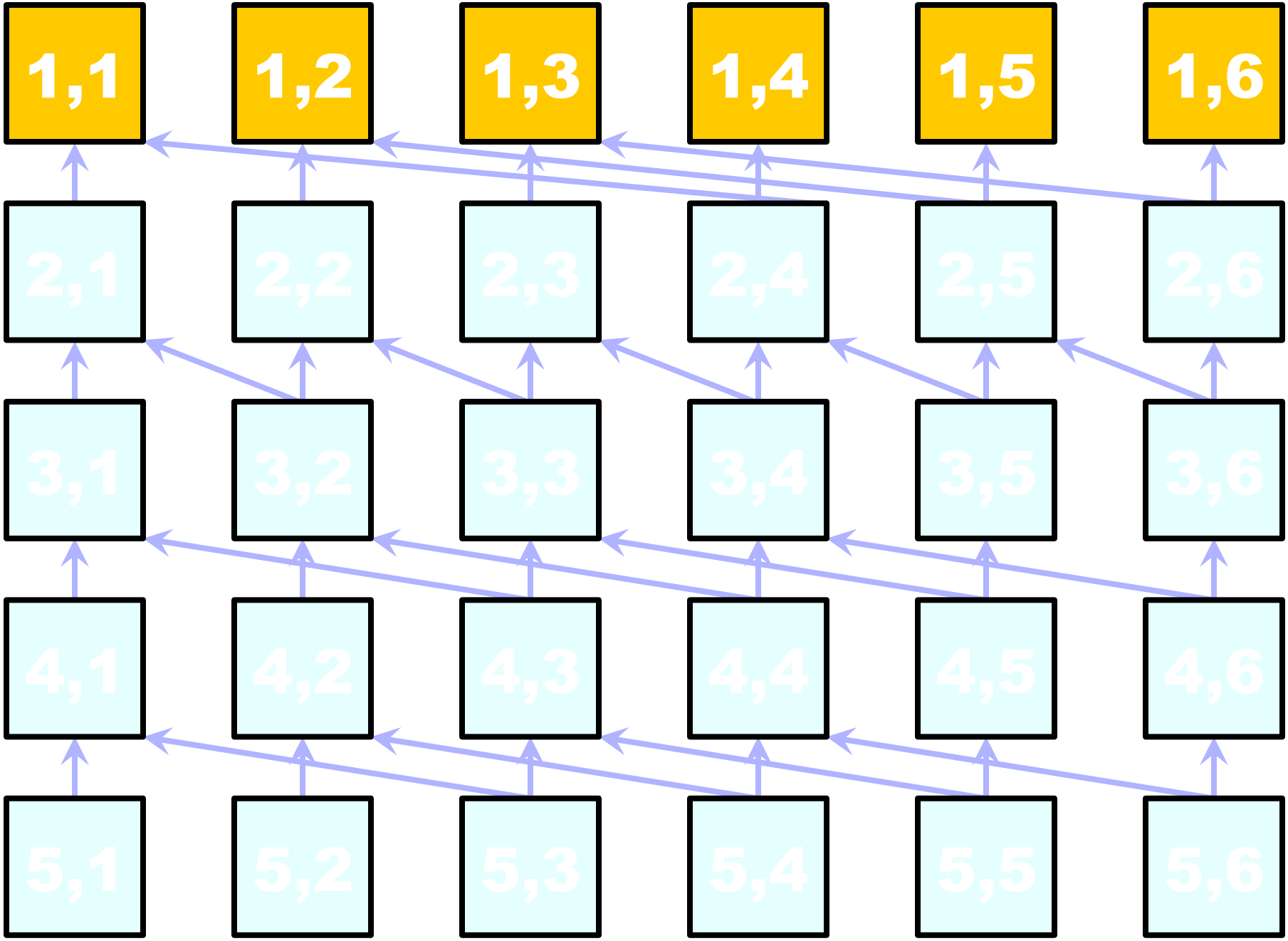
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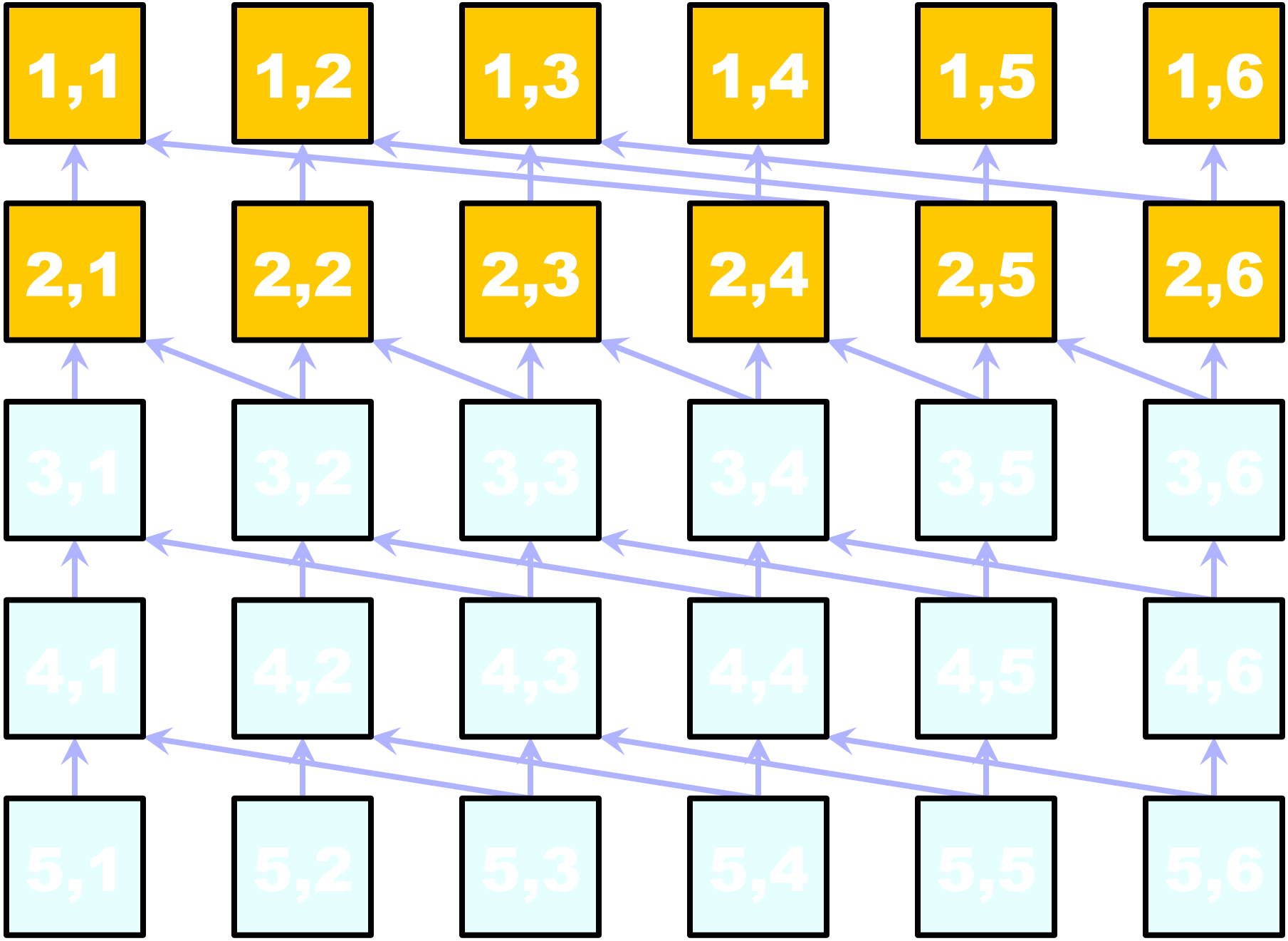
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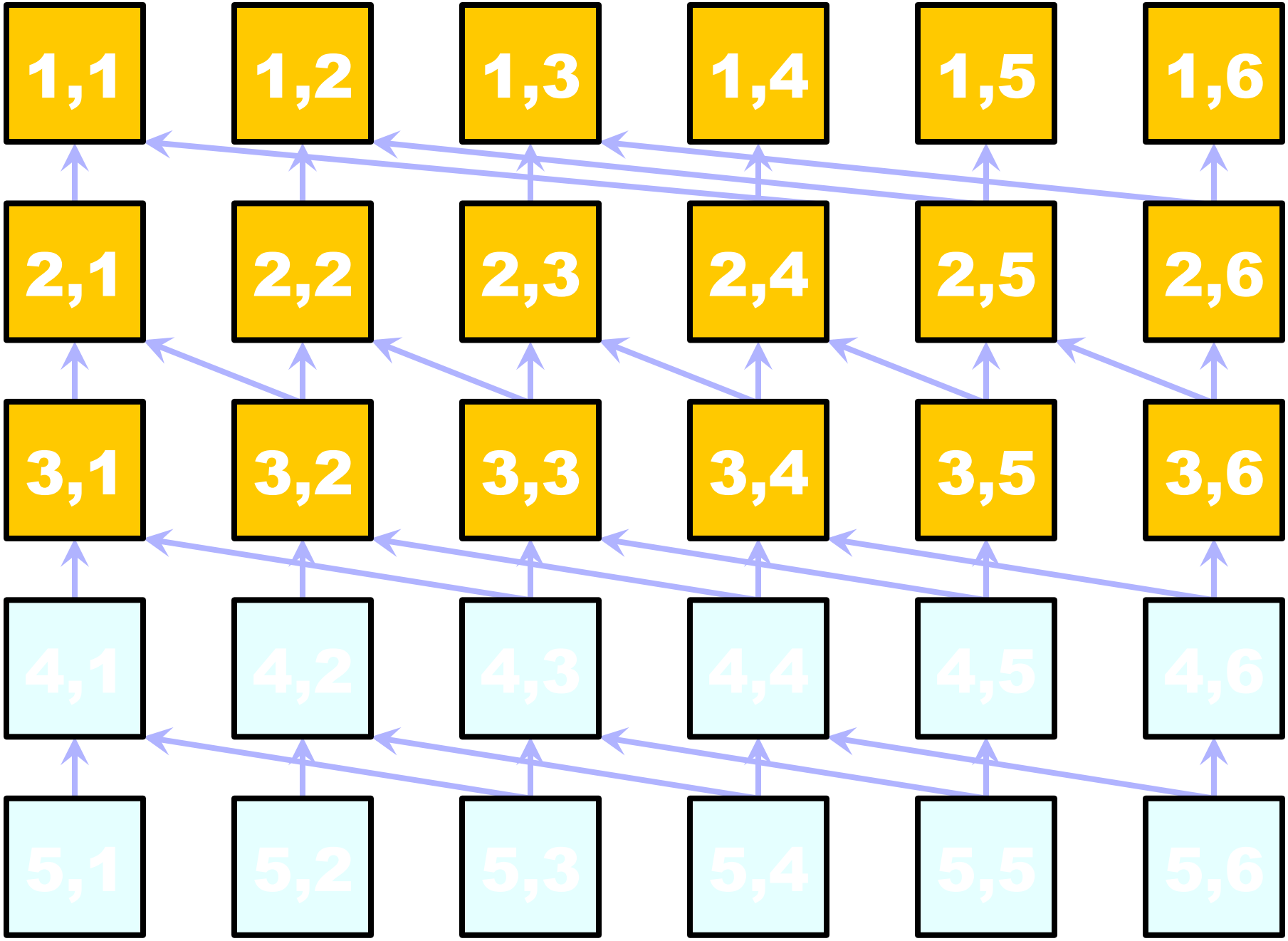
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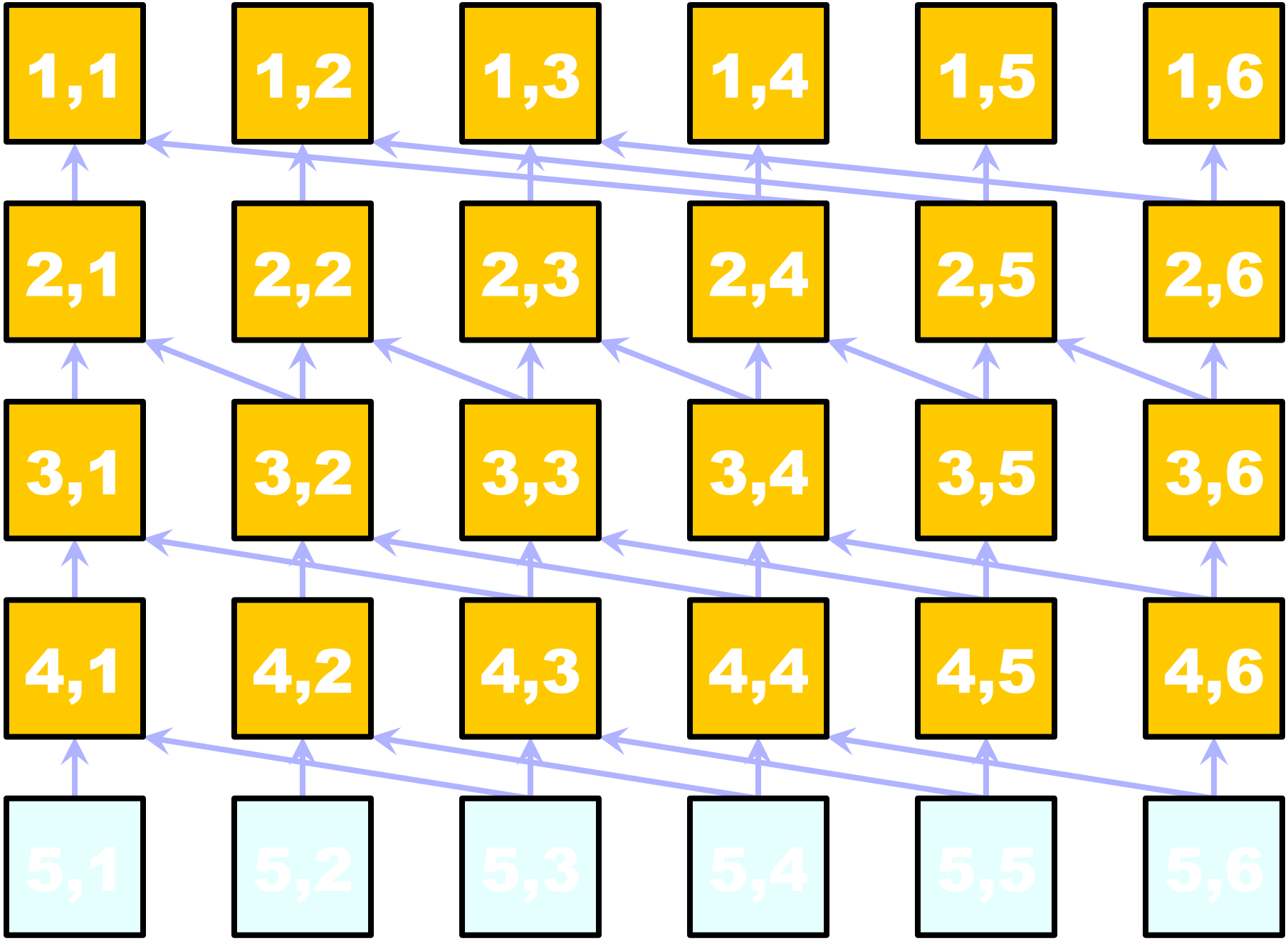
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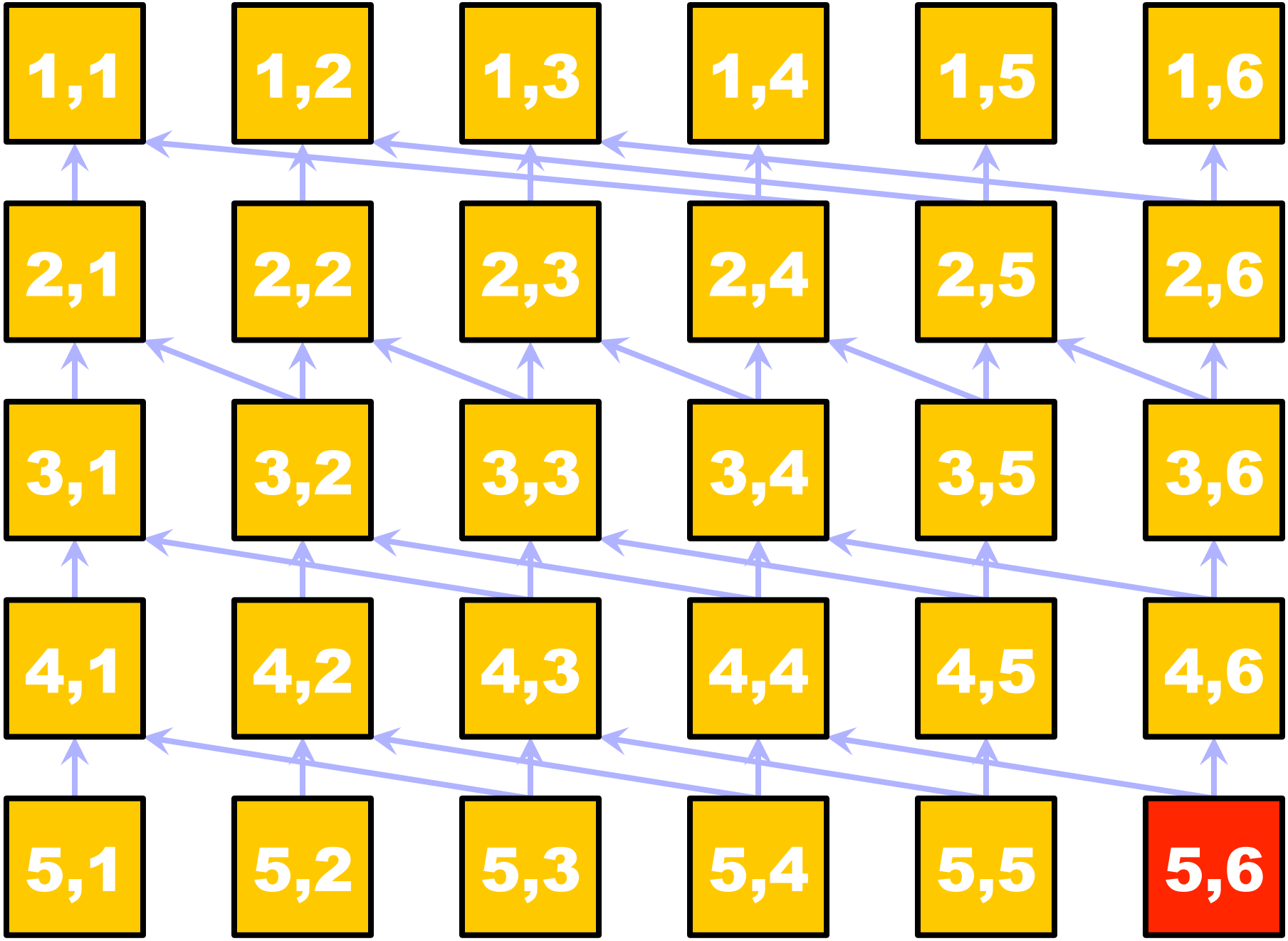
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Question 4

Bottom-Up DP for PARTITION: 2 Rows $\Theta(S)$ space

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3,1	3,2	3,3	3,4	3,5	3,6
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3,1	3,2	3,3	3,4	3,5	3,6
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3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6

5,1	5,2	5,3	5,4	5,5	5,6
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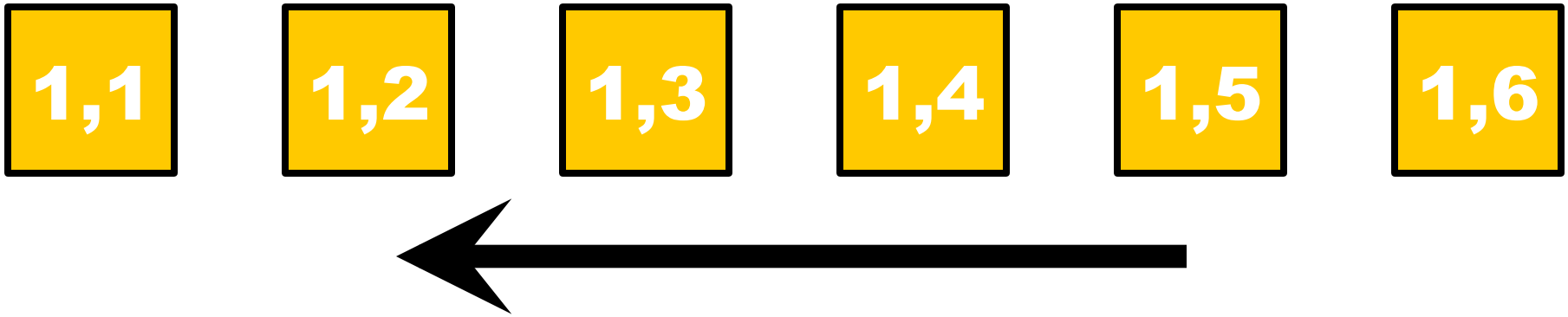
Question 4

Bottom-Up DP for PARTITION: 1 Row $\Theta(S)$ space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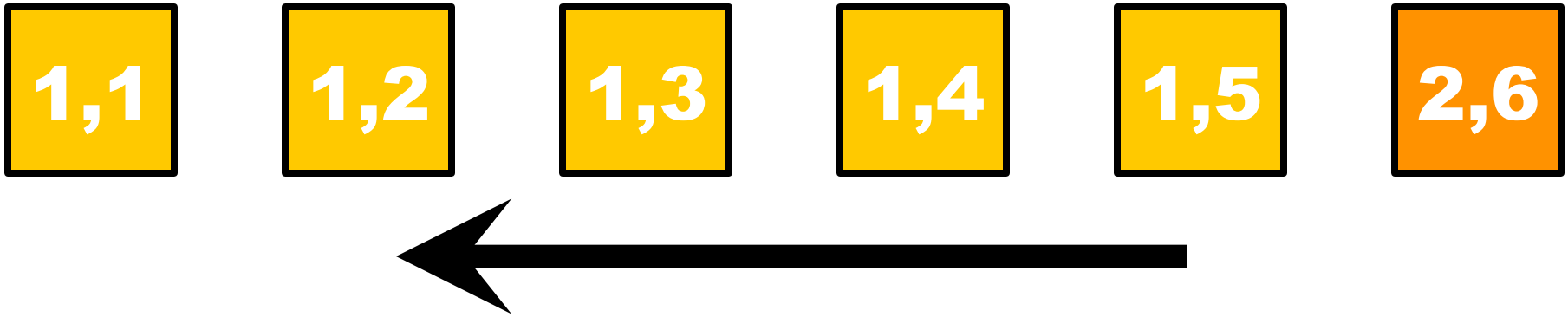
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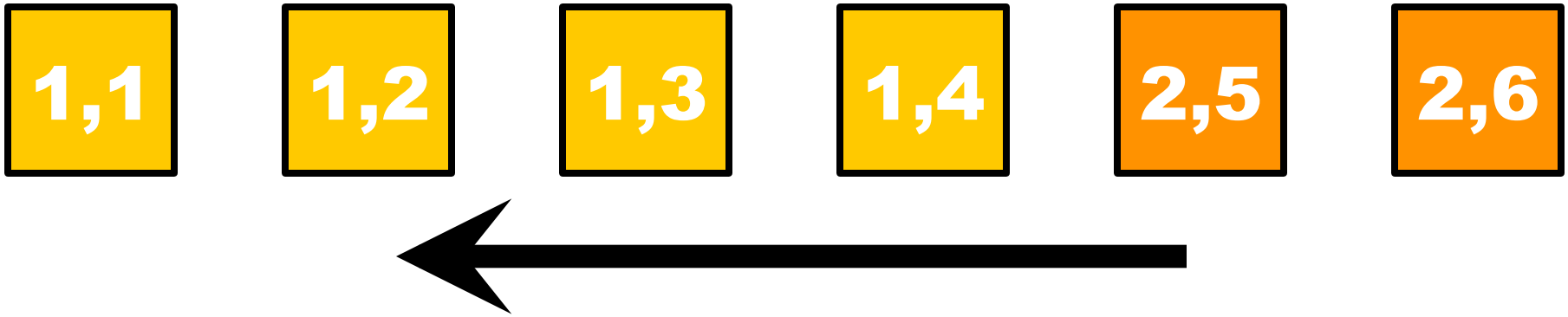
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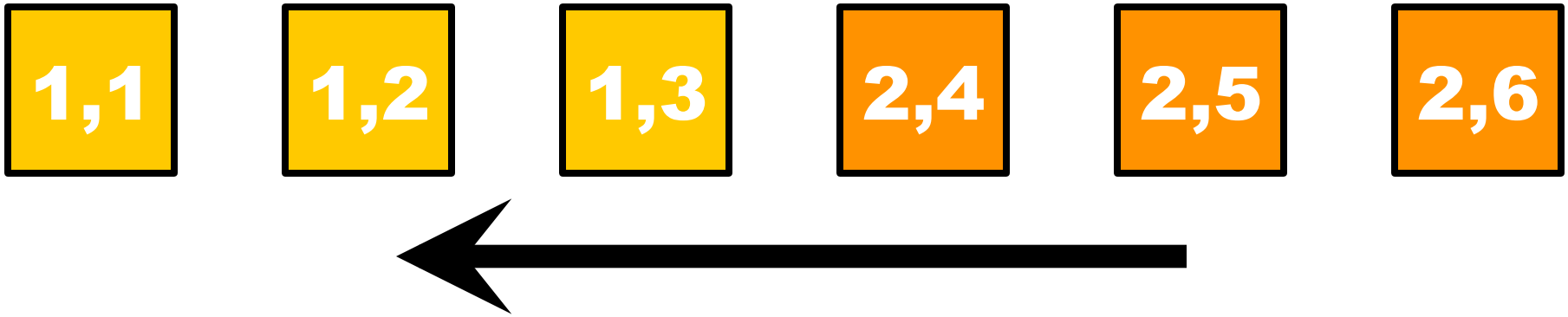
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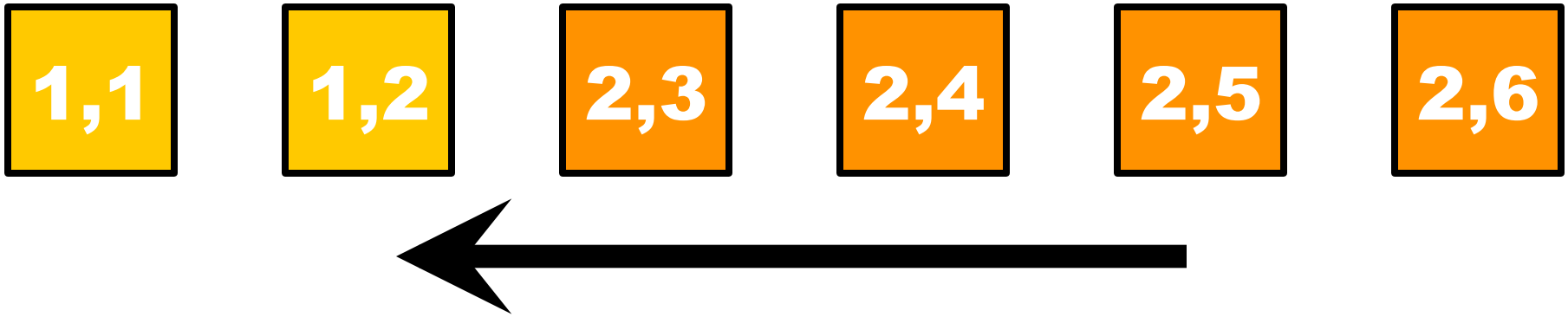
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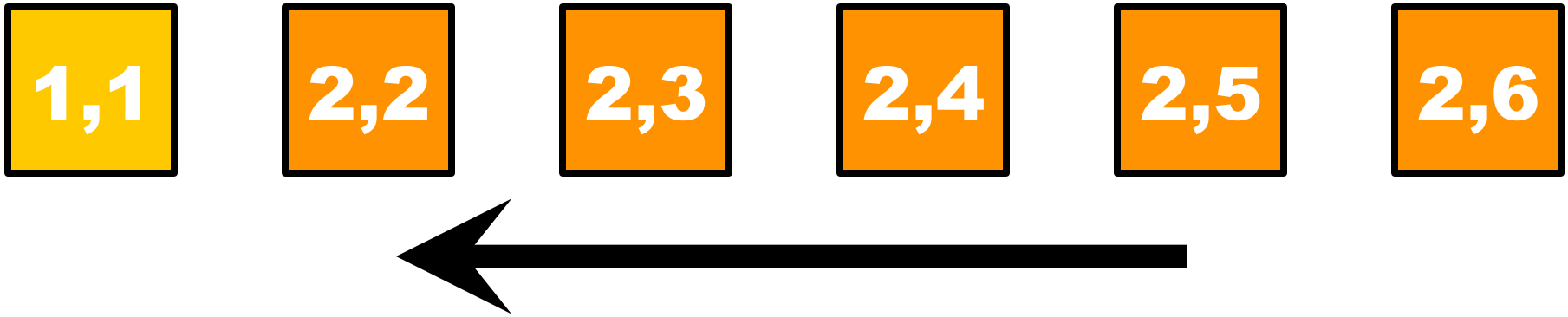
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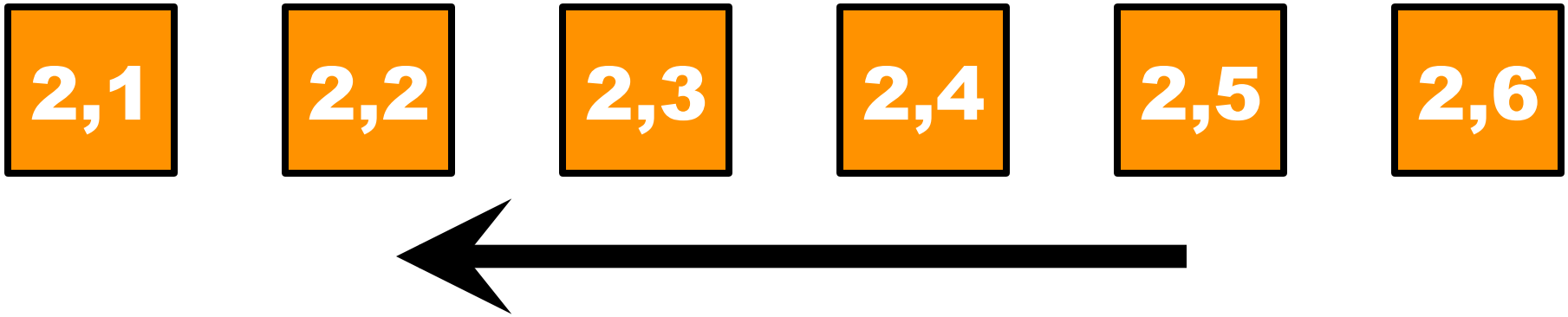
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